MIL-I-16596A
19 March 1969
SUPERSEDING
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15 March 1943

MILITARY SPECIFICATION

INHALATOR, MASK TYPE, OXYGEN ADMINISTRATION

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an inhalation apparatus to be used for the administration of oxygen.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specification and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification:

SPECIFICATIONS

Federal

ZZ-T-831	Tubing; Rubber.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-P-116	Preservation, Methods of.
MIL-T-5542	Thread Compound, Antisieze and Sealing, Oxygen Systems.

FSC 6515

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STANDARDS

Federal

Federal Test Method Standard No. 601	Rubber: Sampling and Testing.

Military

MIL-STD-105	Sampling Procedures for Inspection by Attributes.
MIL-STD-129	Boxes, Set-Up, Paperboard.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC.

UL252	Gas Pressure Regulators
UL404	High-Pressure Gas Gauges

(Underwriters' Laboratories Standards are available from the Underwriters' Laboratories,Inc., Walt Whitman Road, Melville, L.I., N.Y., 11749; 207 East Ohio Street, Chicago, Ill., 60611; and 1655 Scott Boulevard, Santa Clara, Calif., 95050.)

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3. REQUIREMENTS

3.1 Fire and Casualty Hazards.

3.1.1 The contractor shall submit to the contracting agency proof, prior to delivery, that the equipment supplied under this specification conforms to the requirements of Underwriters' Laboratories, Inc., shall be acceptable as evidence that the equipment conforms to this requirement. A copy of each laboratory report shall be furnished to the contracting officer.

3.1.2 In lieu of the label or listing by the Underwriters' Laboratories, Inc., the contractor may submit a laboratory report by a nationally recognized testing agency adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or the issuance of a certificate. The contractor shall submit the laboratory report,which shall include test results and methods used to determine compliance, to the contracting officer for approval. The Government reserves the right to require a facilities report showing the qualifications of the testing laboratory and to conduct a facilities survey to determine its acceptability.

3.1.3 Compliance with the above requirements regarding fire and casualty hazards, does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 Material.

3.2.1 Rubber tubing. Rubber tubing shall be laboratory pure gum, natural rubber, type V, class II, heavy wall tubing, 3/16 $\pm$ 1/64 inch inside diameter, 3/32 $\pm$ 1/64 inch wall thickness, conforming to Federal Specification ZZ-T-831. Color of tubing shall be translucent or black.

3.2.2 Oronasal mask and rebreathing bag. The oronasal mask and rebreathing bag shall be fabricated from natural latex, containing not less than 90 percent rubber hydrocarbon by volume when tested as specified in 4.4.3. Mask and rebreathing bag shall not become brittle, tacky, or acquire a permanent set after subjection to the accelerated aging test specified in 4.4.1. In addition, the tensile strength of the rebreathing bag material shall not decrease by more than 25 percent when subjected to the tensile strength test specified in 4.4.2.

3.2.3 Other materials. All other materials used in the fabrication of components for the inhalator assembly, specified hereinafter, shall be of the best quality, safe, and suitable in all respects for the purpose intended.

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3.3 Style and design. The inhalator shall consist essentially of a lightweight, oxygen pressure regulator, equipped with a pressure gauge and flow indicating device; an oronasal mask and rebreathing bag; and necessary tubing. The inhalator assembly (apparatus) shall be suitable for the safe administration of oxygen.

3.3.1 Regulator with pressure gauge and flow indicating device. Each inhalator apparatus shall be provided with a regulator in accordance with the Defense Medical Purchase Description for Federal Stock Number (FSN) 6680-935-4242, Regulator, Pressure, Medical Gas Administration Apparatus, Lightweight, Oxygen with Flowmeter, in effect on the date of invitation for bids.

3.3.2 Oronasal mask. Each apparatus shall be furnished with one oronasal, inhalator type, rebreathing type mask. Mask shall be light in weight and designed to cover both the nose and mouth of the normal male adult. Material of the mask shall be as specified in 3.2.2. An adjustable head strap shall be provided integral to the mask to retain the mask properly on the face of the patient. The mask shall be designed to accommodate the insertion of a drainage tube without removal of the mask and shall also be provided with an appropriate valve, or valves, as necessary, to permit the mask to function as a rebreathing type. Provisions for connecting the supply tubing and the rebreathing bag to the mask shall be provided. The tubing connector shall be designed to accept tubing of the size as specified in 3.2.1 and 3.3.4 to form a tight, leakproof, connection when fully assembled.

3.3.2.1 Rebreathing bag. The oronasal mask shall be provided with a 500 milliliter or 1000 milliliter capacity rebreathing bag. Material shall be as specified in 3.2.2.

3.3.3 Inspiratory check valve. An inspiratory check valve may be supplied; when supplied, it shall be in a plastic vial. The vial shall have imprinted upon it the following:

"Inspiratory check valve to be inserted in Mask-Bag Connector when prescription calls for NON-REBREATHING. Indications include treatment for poisoning by carbon monoxide and other noxious gases:

1. Remove face piece and rebreathing bag from connector.

2. Pull tail of valve thru connector center hole with tweezer until valve snaps into position.

3. Reassemble mask. Note: Install rebreathing bag on headed end of connector."

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3.3.4 Rubber tubing. Each inhalator apparatus shall be supplied with rubber tubing. Tubing shall be as specified in 3.2.1 and shall be a minimum of 60 inches in length.

3.3.5 Finish. All metal parts of oronasal mask shall be chromium plated or anodized.

3.6 Thread sealants. Organic based thread sealants which are considered to be flammable when in contact with oxygen shall not be used. Thread sealants used for antiseize and sealing purposes shall be teflon thread-sealing tape or thread compounds in strict accordance with Military Specification MIL-T-5542.

3.7 Service data. The contractor shall furnish, with each inhalator apparatus, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Drawings and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. One set of the service data, as indicated above, shall be placed in a sealed, waterproof envelope, and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by Military Standard MIL-STD-129. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request. The service manual shall contain the following precautionary note:

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"CAUTION - DO NOT USE ORGANIC BASED THREAD SEALANTS ON ANY PORTION OF THE REGULATOR. USE ONLY TEFLON THREAD-SEALING TAPE OR 'THREAD COMPOUND, ANTISEIZE AND SEALING, OXYGEN SYSTEMS, 4-OZ TUBE (FSN 8030-243-3284)', AVAILABLE FROM THE GENERAL SERVICES ADMINISTRATION."

3.8 Identification marking. The complete inhalator apparatus, including regulator with pressure gauge and flow indicating device, and the oronasal mask and rebreathing bag shall be clearly and legibly marked with the manufacturer's name or registered trademark.

3.9 Workmanship. Workmanship shall be first class throughout. All components of the inhalator apparatus specified herein shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Military Standard MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be as conducted in accordance with Military Standard MIL-STD-105 and as indicated in table I.

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Table I. Sampling Criteria for Tests in Accordance with MIL-STD-105.

Component	Characteristic	Reqmt Requirement	Test Proceding	Unit of Product	Insp. Level	AQL
Regulator with pressure gauge and flow indicating device	All requirements	SEE FSN 6680-935-4242				
Oronasal Mask	Accelerated Aging Test	3.2.2	4.4.1	1 mask	S-2	*
	Rubber Hydrocarbon Test	3.2.2	4.4.3	1 mask	**	*
Rebreathing Bag	Accelerated Aging Test	3.2.2	4.4.1	1-bag	S-2	*
	Rubber Hydrocarbon Test	3.2.2	4.4.1	1-bag	**	*
	Tensile Strength Test	3.2.2	4.4.2	1-bag	S-2	*
Rubber Tubing	Dimensions	3.2.1 and 3.3.4	ZZ-T-831	1-tube	S-1	*
	Tensile Strength	ZZ-T-831	4.4.2 and ZZ-T-831	1-tube	S-2	*
	Ultimate Elongation	ZZ-T-831	ZZ-T-831	1-tube	S-2	*
	Tensile Strength at 100 percent elongation	ZZ-T-831	ZZ-T-831	1-tube	S-2	*
	Accelerating Aging Test	ZZ-T-831	4.4.1 and ZZ-T-831	1-tube	S-2	*

* Acceptance number shall be zero for all sample sizes.

** Sample size shall be in accordance with applicable test method.

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4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The inhalator apparatus shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accelerated aging test. The oronasal mask, rebreathing bag, and rubber tubing shall be subjected to the accelerated aging test conducted in accordance with method 7221 of Test Method Standard 601.

4.4.2 Tensile strength test. The rebreathing bag and rubber tubing shall be subjected to tensile strength test conducted in accordance with method 4111 of Test Method Standard 601 using appropriate die suitable for performing tensile strength test.

4.4.3 Rubber hydrocarbon by volume. The oronasal mask and rebreathing bag shall be subjected to rubber hydrocarbon test in accordance with method 16111 of Test Method Standard 601.

4.4.4 Other tests. All other tests for rubber tubing shall be in accordance with Federal Specification ZZ-T-831 for Type V, Class II, material. Regulator tests shall be as specified in FSN 6680-935-4242.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One inhalator, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table II should be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Tubing shall be disconnected and formed into a coil of such diameter that it will not kink or take a permanent set. Rebreathing bag shall be rolled in such a manner that it will not be permanently deformed. The mask, tubing, and rebreathing bag, along with the regulator with pressure guage and flow indicating device, shall be separately wrapped in tissue or other suitable material. The wrapped components shall be further packaged in accordance with MIL-P-116, method IA-15. Components shall be arranged and cushioned within the carton in such a manner as to prevent shifting of or damage to the contents.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-601, table III, using type I class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. All metal stitches and other sharp projections on the interior surfaces of the exterior containers shall be covered with tape or other suitable material to prevent rupture of the unit package barrier bag. Closure and strapping shall be in accordance with the appendix of the applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table II, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-621, table I; or PPP-B-636, type CF, class domestic. All metal stitches and other sharp projections on the interior surfaces of the exterior containers shall be covered with tape or other suitable material. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Marking shall be as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurements documents shall specify the following:

a. Title, number and date of this specification.

b. Selections of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification is intended to cover the types that are normally procured to meet military requirements. It does not include all types, sizes, or grades as indicated by the title nor all types which are commercially available.

6.3 The use of asterisks to indicate changes in this specification have been omitted, since the changes are primarily those of format due to the incorporation of modifications of applicable Purchase Descriptions. The most significant change is the use of a single stage oxygen regulator rather than an oxygen-helium regulator. Bidders and contractors are cautioned to evaluate the requirements of this specification based on the entire contents.

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Custodians:
Army - MD
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Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements .

SPECIFICATION
MIL-I-16596A FSN

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMEI [illegible] OO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what wa*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 54, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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